Spatial-Temporal Relation Guided Motion Transfer via Diffusion Model

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***Abstract*—Transferring existing Human-Object Interaction (HOI) motion to novel objects is essential for robotics, virtual reality. Traditional approaches only model spatial surface correspondences between humans and source objects or between source and target objects, ignoring the internal topological structures of humans, the internal topology of objects, the non-surface spatial topological relationships, and temporal motion relations. In this paper, we propose a spatial-temporal relation guided motion transfer framework. Firstly, we define a spatial-temporal relation interaction graph representation(STRIG) to model the human internal topology, object internal topology and human-object global topology together with the temporal motion relation. We propose a STRIGs-guided motion transfer diffusion model for generating spatially, semantically and temporally consistent HOI motions that are adapted to novel objects. To tackle the absence of ground-truth motions after transfer, we introduce a spatial-temporal relation optimization strategy. Extensive experiments demonstrate that our method consistently outperforms other approaches in terms of motion transfer quality, performance, and sequence stability, with particularly robustness under large variations in target object topology.**

***Index Terms*—Human-object interaction, motion transfer, diffusion model.**

I. Introduction

HUMANS interact with objects at every moment. Interaction relationships and motion are at the core of the interactive process and have been widely studied in computer graphics, robotics, and virtual reality. However, generating realistic, physically plausible interaction actions with 3D objects for the agent remains a challenging task due to the complexity of the real world's spatial-temporal relationships. Traditional approaches generate interaction action through manual design or motion capture devices, which are time-consuming and labor-intensive. Transferring the existing Human-Object Interaction (HOI) motion to new target objects offers a more effective paradigm.

Existing methods typically rely on spatial keypoint encoding and shape interpolation as their main approaches. NDF [1] encodes spatial keypoints as SE(3)-equivariant neural descriptor fields, mapping 3D coordinates to continuous descriptors. Tink [2] ensures contact surface consistency through shape interpolation and iterative pose refinement. SSCF [3] combines them by mapping spatial and surface points between source and target objects using a learned correspondence field, thereby optimizing the agent's pose for precise interaction transfer. However, these methods use sparse points or an implicit field as a transfer representation to model the spatial human-object surface correspondence, thereby ignoring the internal topological structures of humans, the internal topology of objects, the non-surface spatial topological relationships, and temporal motion relations, which led to human–object interpenetration and unstable motion sequences.

In this paper, we propose a spatial-temporal relation guided motion transfer framework, which substantially improves the quality, performance, and similarity of HOI motion generation. Firstly, we introduce a spatial-temporal relation interaction graph representation (STRIG) to model the human internal topology, object internal topology, and human-object global topology together with the temporal motion relation. STRIG is a heterogeneous graph, which contains a human sub-graph, an object sub-graph, and HOI edges. Nodes and edges in the graph and subgraph encode the spatial, semantic and temporal information. To generate spatially, semantically and temporally consistent HOI motions, we propose a STRIGs-guided motion transfer diffusion model that learns the implicit source-target object correspondence and establishes a mapping between source and target human motions. To address the issue of the absence of ground-truth motions after transfer, we introduce a spatial-temporal relation optimization strategy that ensures both source and target motion similarity and physical plausibility. Experiments demonstrate that our method outperforms state-of-the-art approaches in motion transfer quality, efficiency, and sequence stability. As shown in Fig. 1, our framework successfully transfers human–object interactions across objects with distinct geometries - adapting the arm placement to fit the low armrests of a compact office chair. These results demonstrate

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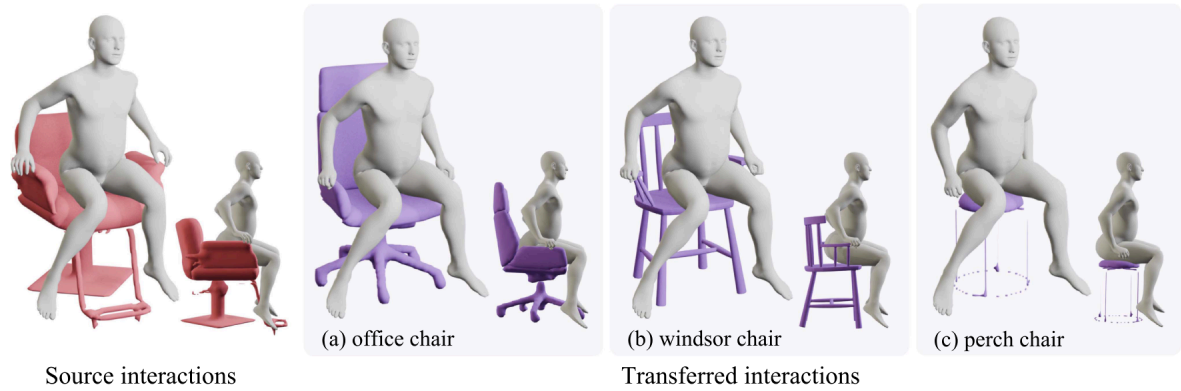


Fig. 1. Our spatial-temporal relation guided motion transfer framework effectively transfer human–object interactions to diverse target objects. Starting from a source interaction (pink), the target transferred motions (purple) align with objects of varying geometry, such as the lower armrests of a compact office chair, while maintaining motion similarity and physical plausibility for windsor and perch chairs, which exhibit significant geometric variations, producing collision-free and semantically consistent interactions.

the robustness of our method under large target object topology variations.

In summary, our main contributions are as follows:

- We propose a spatial-temporal relation guided motion transfer framework, which substantially improves the quality, performance, and stability of HOI motion generation.
- We define a spatial-temporal relation interaction graph representation (STRIG) that encodes the human internal topology, object internal topology, human-object global topology together with the temporal motion relation.
- We propose a STRIGs-guided motion transfer diffusion model, enabling high-quality motion transfer based on implicit source–target correspondences. Moreover, we introduce a spatial-temporal relation optimization strategy that addresses the lack of ground-truth supervision after transfer and ensures the physical plausibility of the generated motions.

II. Related Work

In this section, we review two types of research related to our work: HOI motion transfer and HOI motion diffusion. For a more comprehensive overview of prior works, we recommend readers refer to these reviews [4], [5], [6].

A. HOI Motion Transfer

HOI motion transfer refers to the process of adapting a human's interaction motion from one object to another, while maintaining consistent spatial and temporal relationships between the human and the novel object. Unlike traditional motion retargeting, which focuses on transferring motion between characters with different skeletons [7], [8], [9], [10], shapes [9], [10], [11], or scales [12], [13], HOI motion transfer keeps the human agent fixed and varies only the interacting object. The goal is to generate semantically coherent and physically plausible motions that preserve both the intent and spatial integrity of the original interaction. Existing methods for HOI interaction transfer typically rely on an interaction representation, which is then optimized to facilitate the transfer.

From the perspective of establishing the mapping between objects, surface contact correspondences typically refer to the regions on the object's surface that are in close proximity to the human agent during interaction. One common approach is contact point optimization [2], [14], [15], [16], which ensures that these contact points either remain in contact or retain the same relative distances to the agent as in the source interaction, even when transferring the interaction to a new object. However, these methods are highly sensitive to the accuracy of the surface contact correspondence. When there are no direct contact points between the character's motion sequence and the object, maintaining spatial consistency and physical plausibility becomes significantly more challenging. To overcome this limitation, spatial point correspondences define consistent relationships between corresponding points in the 3D spaces surrounding two objects [1], [17], [18]. By maintaining the relative positions of key points, these methods ensure that motion is transferred between objects, even in the absence of direct contact points. While offering precise localization and accurate mappings of 3D spaces, spatial point methods encounter difficulties when dealing with target objects that have incomplete or partial observations, as they depend on fully detailed surface data for accurate mapping.

More recently, the SSCF [3] leverages implicit deformation field to combine both spatial and surface correspondences, improving interaction transfer between objects by capturing spatial surface relationships. However, SSCF ignore the internal topological structures of humans, the internal topology of objects, the non-surface spatial topological relationships, and temporal motion relation. These make the transfer highly susceptible to motion drift, semantic misalignment, and physical inconsistency, especially when the target object exhibits significant topological or geometric variations. In this paper, we introduce a graph representation that models the internal, global spatial and temporal relation, ensuring motion transfer quality, performance, and sequence stability.

B. HOI Motion Diffusion

Diffusion models has attracted significant attention in the field of natural language processing, computer vision, computer graphic for its high accuracy and generalization [19], [20], [21], [22], [23], [24], [25], [26]. These models [27], [28] emulate

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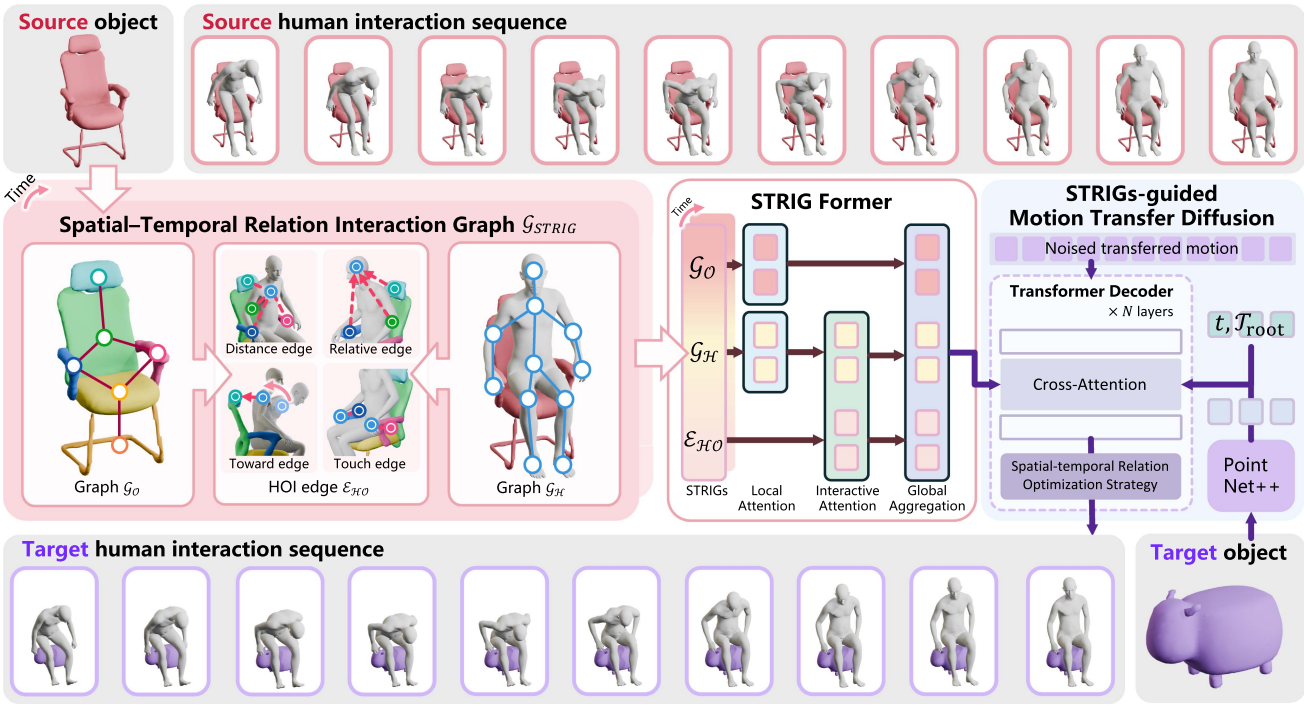


Fig. 2. Overview of the spatial-temporal relation guided motion transfer framework. We first encode the source interaction with our STRIG representation. Then, we apply a STRIGs-guided motion transfer diffusion model to transfer the source human interaction sequence to the target object. Next, we introduce a spatial-temporal relation optimization strategies to regularize the diffusion process.

thermodynamic diffusion by progressively adding noise to data and training a neural network to reverse this process, thereby generating realistic samples through iterative denoising from pure noise. MotionDiffuse [29], MDM [30] and Flame [31] concurrently introduced the diffusion paradigm into the domain of motion generation.

Building on these advances, diffusion-based methods have recently enabled the generation of interactive human motions conditioned on multi-modal inputs, including images [32], [33], [34], text [35], [36], [37], and video [38], [39], [40]. However, such approaches typically rely on meticulously curated HOI datasets to synthesize either static poses [35] or dynamic motion sequences [41], [42], [43], [44]. In HOI motion transfer, curated paired samples for motion translation are generally unavailable. Consequently, most existing methods [3], [14] depend on parameter optimization, which limits the physical plausibility of the generated motions. To address this challenge, we propose a spatial relation optimization strategy that compensates for the absence of ground-truth supervision. In this paper, we introduce a STRIGs-guided motion transfer diffusion model, which leverages a spatial–temporal relation interaction graph to capture implicit spatial and temporal relation, enabling high-quality and sequence stable motion transfer.

III. Method

Given a source object, corresponding HOI motion sequence and a target object as input, our method outputs target transferred motion sequence with the novel target object. The pipeline is shown in Fig. 2. We first represent the source object and HOI motion sequence as Spatial–Temporal Relation Interaction Graphs (STRIGs), which encode the human internal topology, object internal topology, and human-object global topology together with the temporal motion relation (Section III-A). Then, we design a STRIGs-guided motion transfer diffusion model which adapts the STRIGs suitable for source object to the target object (Sections III-B and III-C). To ensure reliable generation without ground-truth transferred motions, we introduce a set of spatial relation optimization strategies that regularize the diffusion process.(Section III-D)

A. Spatial-Temporal Relation Interaction Graph

We define the object point cloud as $\mathcal{O}$ and the corresponding human interaction sequence as $\mathcal{H}(\mathcal{T},\mathcal{R}) = \{h_t \mid t = 0,\ldots,T\}$, where $\mathcal{H}$ is parameterized by the rigid transformation $\mathcal{T}$ and the human joint rotation angles $\mathcal{R}$. Given a source interaction $(\mathcal{H}^{\mathrm{src}}, \mathcal{O}^{\mathrm{src}})$ and a target object $\mathcal{O}^{\mathrm{tgt}}$, our objective is to derive the human interaction sequence $\mathcal{H}^{\mathrm{tgt}}$ for the target object, ensuring the consistency of the HOI motion transfer by learning the spatial and temporal relation across the two domains.

We propose a Spatial–Temporal Relation Interaction Graph (STRIG) representation to effectively encode the human internal topology, object internal topology, and human-object global topology together with the temporal motion relation. STRIG is defined as a heterogeneous graph with (1).

$$\mathcal{G}_{STRIG} = \Big\langle \underbrace{(\mathcal{V}_{\mathcal{H}}, \mathcal{E}_{\mathcal{HH}}, \mathcal{X}_{\mathcal{H}})}_{\mathcal{G}_{\mathcal{H}}}, \underbrace{(\mathcal{V}_{\mathcal{O}}, \mathcal{E}_{\mathcal{OO}}, \mathcal{X}_{\mathcal{O}})}_{\mathcal{G}_{\mathcal{O}}}, \mathcal{E}_{\mathcal{HO}}, \mathcal{Y} \Big\rangle \quad (1)$$

where $\mathcal{G}_{\mathcal{H}}$, $\mathcal{G}_{\mathcal{O}}$ and $\mathcal{E}_{\mathcal{HO}}$ denote the human graph, object graph and HOI edge, respectively.

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$\mathcal{G}_{\mathcal{H}}$ contains nodes $\mathcal{V}_{\mathcal{H}}$ and edges $\mathcal{E}_{\mathcal{H}\mathcal{H}}$ and is constructed based on the SMPLX skeletal topology [45]. Each node encodes joints-level spatial and temporal information, including joint spatial position, 6D rotation [46], motion state (static or dynamic) and movement direction.

The object subgraph $\mathcal{G}_{\mathcal{O}}$ contains nodes $\mathcal{V}_{\mathcal{O}}$ and edges $\mathcal{E}_{\mathcal{O}\mathcal{O}}$. We employ the PointNet++ [47] to construct them by decomposing the raw object point cloud into distinct geometric parts. For CHAIRS dataset [48], we utilize provided fine-grained part annotations. Each geometric part is then associated with an axis-aligned bounding box. The internal object edges $\mathcal{E}_{\mathcal{O}\mathcal{O}}$ are established based entirely on the geometric overlap relations among these bounding boxes to capture the structural integrity of the object. Correspondingly, the human-object interaction edges ($\mathcal{E}_{\mathcal{H}\mathcal{O}}$) in (2) are computed dynamically using the spatial distances, threshold-based contacts, and motion trajectories relative to these unlabeled geometric parts. The node feature is then enriched by fusing latent features extracted via the unsupervised pretrained encoder ULIP [49] with explicit attributes such as the part centroid and bounding box geometry. Edges $\mathcal{E}_{\mathcal{O}\mathcal{O}}$ are constructed based on bounding box overlap relations, encoding geometric constraints among object parts and maintaining the structural integrity of the object subgraph.

HOI edge $\mathcal{E}_{\mathcal{H}\mathcal{O}}$ represents the interaction between human and objects as (2).

$$\mathcal{E}_{\mathcal{H}\mathcal{O}} = \{\mathcal{E}_{dist}, \mathcal{E}_{rel}, \mathcal{E}_{touch}, \mathcal{E}_{tow}\} \tag{2}$$

There are 4 types of the interaction: 1) $\mathcal{E}_{dist}$: the shortest distance between human joints and object part point clouds; 2) $\mathcal{E}_{rel}$: the relative position of human joints with respect to the object bounding boxes in a spatial–temporal context; 3) $\mathcal{E}_{touch}$: contact relationships when the distance falls below a predefined threshold; 4) $\mathcal{E}_{tow}$: whether the joint moves toward a specific object part.

Graph embedding $\mathcal{X}_{\mathcal{H}}$ and $\mathcal{X}_{\mathcal{O}}$ contains the corresponding node and edge embeddings, integrating geometric, temporal and semantic attributes. Graph label $\mathcal{Y}$ is defined by the global human–object bounding volume for each frame graph $\mathcal{G}_{\text{STRIG}}$, serving as a compact descriptor of the holistic interaction state.

B. STRIGs-Guided Motion Transfer Diffusion

We propose a STRIGs-guided motion transfer diffusion model that establishes implicit source-target correspondences and generates spatially and temporally consistent target interaction motion sequence, as shown in Fig. 2.

The diffusion model describes a forward Markov process $q(x_t|x_{t-1}) = \mathcal{N}(\sqrt{\alpha_t}x_{t-1}, (1-\alpha_t)I)$, which progressively adds Gaussian noise to a clean data sample $x_0^{\mathtt{src}} \sim p_{\mathcal{H}^{\mathtt{src}}}(x_0^{\mathtt{src}})$ from the source human interaction sequences $\mathcal{H}^{\mathtt{src}}$. In the reverse process, the model learns to denoise a sample from this prior distribution, gradually recovering the clean data x_0. Instead of predicting ϵ_t by DDPM [28], we follow the CLoSD [50], using an equivalent formulation to predict the target human interaction sequence $\hat{x}_0^{\mathtt{tgt}} \sim p_{\mathcal{H}^{\mathtt{tgt}}}(x_0^{\mathtt{tgt}})$, making it similar to $x_0^{\mathtt{src}}$. The objective function is formulated as (3).

$$\mathcal{L} = \mathbb{E}_{t\sim[0,T]} STRoptim(x_0^{\mathtt{src}}, \hat{x}_0^{\mathtt{tgt}}) \tag{3}$$

where predicting $\hat{x}_0^{\mathtt{tgt}}$ leads to improved results for the target human interaction sequence $\mathcal{H}^{\mathtt{tgt}}$. To address the distribution mismatch between the predicted $\hat{x}_0^{\mathtt{tgt}}$ and the clean data $x_0^{\mathtt{src}}$, we introduce a spatial-temporal relation optimization strategy (Section III-D) to apply a target loss as a geometric loss at each denoising step, ensuring motion transfer similarity and physical plausibility.

The model is designed with a transformer decoder backbone [51] and is conditioned on: 1) STRIGs latent embeddings, extracted from the STRIGs Former encoder (Section III-C). 2) The source global SMPL-X root transformation $\mathcal{T}_{\text{root}} \in SE(3)$ at the first frame, constructed as [52] providing the model with the initial global position and orientation of the source body pose. 3) The geometry of the target object, embedding through the PointNet++ [47], which is trained simultaneously with the diffusion model. At each step $t \in [0, T]$, the noisy motion transfer prediction $\hat{x}_t^{\mathtt{tgt}}$ is combined with standard positional embeddings to encode the token indices, and is input to the backbone as a single input token. Simultaneously, the model's conditions are constructed and injected through the cross-attention blocks of each of the transformer's layers.

To inject the aforementioned conditions into the transformer decoder, we adopt a concatenation strategy. The Query is exclusively provided by the noised motion prediction $\hat{x}_t^{tgt}$ with a sequence length of T. The Key and Value are computed from a unified memory sequence formed by concatenating all condition tokens along the sequence dimension. Specifically, the memory sequence comprises: the diffusion timestep (1 token), the frame-level STRIG embeddings (T tokens), the initial root transformation (1 token), and the target object point embeddings (one token per sampled vertex). To allow the cross-attention mechanism to distinguish between these disparate modalities, a distinct learnable type embedding is added to each condition token prior to concatenation.

C. STRIG Former

To extract the HOI global structure and spatial and temporal relation embedded in STRIGs, we introduce STRIG Former, a multi-stage extraction block, which transforms STRIGs into sequential latent embeddings.

1) Local Attention: We apply a graph attention mechanism [53] forcing the subgraph's nodes focusing on relative spatial local features. Specifically, we update the node information in two subgraphs such that the nodes focus on the information from neighboring nodes during the update, thereby obtaining local attention features. Moreover, we map the nodes from different human subgraph $\mathcal{G}_{\mathcal{H}}$ and object subgraph $\mathcal{G}_{\mathcal{O}}$ to the same latent domain with (4).

$$\mathbf{n}_i' = \sum_{j\in\mathcal{N}(i)\cup\{i\}} \alpha_{i,j}\mathbf{\Theta}_t\mathbf{n}_j, \mathbf{n} \in \mathcal{G}_{\mathcal{H}}, \mathcal{G}_{\mathcal{O}} \tag{4}$$

where $\alpha_{i,j}$ are the attention coefficients, and $\mathbf{\Theta}_t$ are learnable MLP. The node i is updated based on the weighted aggregation of its neighboring node $\mathcal{N}(i)$. The local attention coefficients

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$\alpha_{i,j}$ are computed based on (5).

$$\alpha_{i,j} = \frac{\exp\left(\sigma\left(\mathbf{a}_s^\top \mathbf{\Theta}_s \mathbf{n}_i + \mathbf{a}_t^\top \mathbf{\Theta}_t \mathbf{n}_j\right)\right)}{\sum_{k\in(\mathcal{N}(i)\cup\{i\})} \exp\left(\sigma\left(\mathbf{a}_s^\top \mathbf{\Theta}_s \mathbf{n}_i + \mathbf{a}_t^\top \mathbf{\Theta}_t \mathbf{n}_k\right)\right)} \tag{5}$$

2) Interactive Attention: We design interactive attention to capture the spatial-temporal interact relation of $\mathcal{E}_{\mathcal{HO}}$ in STRIG based on multi-head dot product attention [54].

The node feature is updated by weighted aggregation of its own features and edge features as (6).

$$\mathbf{n}_i' = \mathbf{W}_1 \mathbf{n}_i + \sum_{j\in\mathcal{N}(i)} \alpha_{i,j} \mathbf{W}_2 \mathbf{e}_{ij}, \mathbf{e} \in \mathcal{E}_{\mathcal{HO}}, \mathbf{n} \in \mathcal{G}_{\mathcal{H}} \tag{6}$$

where $\mathbf{e}$ represents the interaction edge features, and $\mathbf{W}$ denotes learnable parameters. The interactive attention coefficients $\alpha_{i,j}$ are computed with (7).

$$\alpha_{i,j} = \text{softmax}\left(\frac{(\mathbf{W}_3 \mathbf{n}_i)^\top (\mathbf{W}_2 \mathbf{e}_{ij})}{\sqrt{d}}\right) \tag{7}$$

3) Global Aggregation: We aggregate feature across the heterogeneous graph and generate global feature. Different aggregation methods are applied based on the edge types. For the human subgraph edges $\mathcal{E}_{\mathcal{HH}}$, the feature is updated centered around its nodes, and local attention is still used for updating nodes within the subgraph. For the interaction edges $\mathcal{E}_{\mathcal{HO}}$, the update between subgraphs is computed with (8) [55]. Specifically, the aggregation compresses the node dimension while strictly preserving the temporal dimension, resulting in exactly T frame-level tokens. This frame-level granularity ensures temporal alignment with the motion sequence during cross-attention.

$$\mathbf{n}_i' = \mathbf{W}_1 \mathbf{n}_i + \mathbf{W}_2 \cdot \text{mean}_{j\in\mathcal{E}_{\mathcal{HO}}(i)} \mathbf{o}_j, \mathbf{n} \in \mathcal{G}_{\mathcal{H}}, \mathbf{o} \in \mathcal{G}_{\mathcal{O}} \tag{8}$$

The Jumping Knowledge Aggregation [56] is applied based on concatenation to aggregate the node features of $\mathcal{G}_{\mathcal{H}}$ obtained at each stage.

In summary, the human subgraph $\mathcal{G}_{\mathcal{H}}$ serves as the active motion carrier, while the object graph $\mathcal{G}_{\mathcal{O}}$ mainly provides static geometric and structural context. Fine-grained dynamic interaction cues, such as relative joint-part relations, contact states, and motion directions, are encoded in the attributed HOI edges $\mathcal{E}_{\mathcal{HO}}$. The interactive attention is designed to update human nodes by attending to these dynamic edge features. The structural context of $\mathcal{G}_{\mathcal{O}}$ is then integrated at the global aggregation stage.

D. Spatial-Temporal Relation Optimization Strategy:

To address the lack of ground-truth after transfer and generate physical plausible motions, we introduce a spatial relation optimization strategy to evaluate the similarity between $\mathcal{H}^{\mathtt{src}}$ and $\mathcal{H}^{\mathtt{tgt}}$ with four terms.

1) Pose Similarity Loss: We introduce a pose similarity loss to enhance the similarity of the overall body pose and spatial-temporal pose with (9).

$$\mathcal{L}_{SIM} = \lambda_{agl}\mathcal{L}_{agl} + \lambda_{rel}\mathcal{L}_{rel} + \lambda_{vel}\mathcal{L}_{vel} \tag{9}$$

The pose similarity loss contains three components: 1) joint-rotation term $\mathcal{L}_{agl}$ that aligns 6D rotations across all joints, 2) relative-position term $\mathcal{L}_{rel}$ that preserves body spatial relations in root-centered coordinates (using the pelvis for the body), and 3) velocity term $\mathcal{L}_{vel}$ that matches transfer-joint offsets to ensure smoothness.

2) Interaction Loss: We propose an interaction loss to enhance physical realism and ensure stable human– object contact by keeping foot–ground consistency and stabilizes global body placement with (10).

$$\mathcal{L}_{INT} = \lambda_{sdf}\mathcal{L}_{sdf} + \lambda_{plv}\mathcal{L}_{plv} + \lambda_{foot}\mathcal{L}_{foot} \tag{10}$$

The interaction loss also contains three components: 1) SDF consistency term $\mathcal{L}_{sdf}$ aligns the signed distance fields (SDFs) of source and target human meshes to preserve inter-object clearances, 2) pelvis-position term $\mathcal{L}_{plv}$ anchors the pelvis position and SDF to maintain physically consistent grounding, 3) foot consistency term $\mathcal{L}_{foot}$ enforces stable and realistic foot–ground interactions with (11).

$$\mathcal{L}_{foot} = w_{vel}\mathcal{L}_{vel} + w_y\mathcal{L}_y + w_{pos}\mathcal{L}_{pos} + w_{hgt}\mathcal{L}_{hgt} \tag{11}$$

where $\mathcal{L}_{vel}$ penalizes foot velocity during contact, $\mathcal{L}_y$ reduces vertical jitter, $\mathcal{L}_{pos}$ enforces spatial stability, and $\mathcal{L}_{hgt}$ constrains the foot height within a narrow range around the ground plane. The contact mask is computed from the source sequence ($y < \tau$), ensuring that the constraint is applied only when the source foot is in contact with the ground.

3) Penetration Loss: We introduce a bidirectional penetration loss to penalize HOI interpenetration and enforce a safety margin between human and objects. For two watertight meshes, M_h (human) and M_o (object), we compute the SDF of each mesh relative to the other and apply a hinge penalty when the distance is below a margin m. The bidirectional penetration loss is formulated with (12).

$$\mathcal{L}_{\text{PEN}} = \mathbb{E}[\text{ReLU}(m - \text{SDF}_{h\leftarrow o})] + \mathbb{E}[\text{ReLU}(m - \text{SDF}_{o\leftarrow h})] \tag{12}$$

where $\text{SDF}_{h\leftarrow o}$ and $\text{SDF}_{o\leftarrow h}$ are the signed distances from object vertices to the human surface, and vice versa. The ReLU ensures that only distances within the 1 mm margin contribute to the loss, maintaining a minimal clearance between meshes.

IV. Experiments

A. Experimental Settings

1) Dataset: We primarily focus on human–chair interactions that encode rich relative spatial information. For the CHAIRS [48], we sample continuous 10 frames as the source interactions sequence based on the interaction instruction for each sequence in the dataset. For the target objects, we randomly select ShapeNet [57] objects that afford chair-like interactions and pair each source sequence with a target object, yielding 50 K training pairs. We also use the CHAIRS and PARK[1] data as target objects for HOI motion transfer to validate the generalization of our method. CHAIRS and PARK data provides a diverse set of shapes and functionalities and enabling high-quality, semantically consistent motion transfer. During training, the target

[1]Data available at https://assetstore.unity.com/packages/3~d/environments/playground-low-poly-191533.

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Fig. 3. Static transfer results. Our method transfers source interactions (pink) to target objects (purple) with diverse geometries while maintaining semantic consistency and physical plausibility.

objects are further subjected to random scaling within $\pm 10\%$ to enhance robustness and prevent overfitting.

2) Implementation Details:: The STRIGs-guided Motion Transfer Diffusion model is implemented as a 4 layer Transformer decoder with a latent dimension of 512 and 4 attention heads. The GELU activation function is applied throughout the decoder. For point encoding, we adopt the PointNet++ architecture [47] with a latent dimension of 1024, taking as input 2048 randomly sampled vertices with surface normals from the target object. In the STRIG Former, the attention module employs 4 graph attention heads. The model is trained under the DDPM [28] with a cosine noise schedule, while DDIM [58] is used during inference. Both training and inference are performed with 10 diffusion steps. We employ the AdamW [59] optimizer with a learning rate of 1×10^{-4} for noise optimization and apply an exponential moving average decay rate [60] of 0.999. Training is conducted for 25 K iterations using $8\times$ NVIDIA A6000 GPUs. The loss coefficients are set as follows: all $\lambda_i = 1 \times 10^3$ and all $w_i = 1.0$.

B. Qualitative Results

1) Qualitative Results Analysis:: Fig. 3 presents the qualitative results of static HOI motion transfer using our method. The red HOI pairs represent the source interactions, while the purple pairs correspond to the transferred target interactions. Each source interaction is transferred to four target objects with distinct geometries and topologies. The target objects are selected from ShapeNet (top two rows), CHAIRS (third row), and PARK (fourth and fifth rows), covering a wide range of structural variations. Our method effectively preserves pose consistency while adapting to target objects with different structures. Despite substantial geometric differences, the transferred results maintain coherent spatial relationship between the human and the interacting object. For example, the contact between the buttocks and the seat is accurately retained across various chair shapes, ensuring both semantic and physical plausibility. For the "Scratch a little in the chair." pose [48] (first row, right), the transferred results consistently preserve the grasping behavior, with the right hand remaining attached to the chair edge. In cases "Sit with your hands on the armrests." where the source object has armrests but the target does not (second row, left), the arm naturally adjusts downward, maintaining motion similarity and physical realism. The third row "Change to another pose of sitting." and "Change the pose of your legs." further demonstrates that our method captures the underlying force-support interaction relationships between humans and objects, adapting well regardless of backrest presence or differences in size and height. To test generalization beyond chairs, we apply the model to objects from the PARK category, such as ladders, swings, and seesaws (last two columns). Fig. 3 demonstrates that the transferred results remain semantically consistent and physically plausible, demonstrating the robustness of our method, even under large geometric and topological variations.

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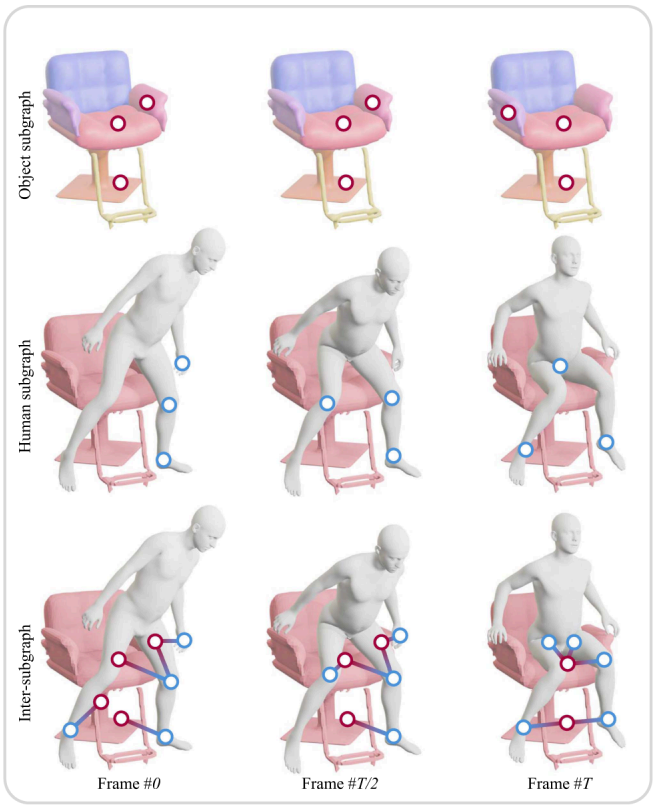


Fig. 4. STRIG Former attention visualization. The top two rows visualize the top-3 highest attention nodes within the object subgraph (red nodes, located at the object part bounding box centers) and the human subgraph (blue nodes, indicating joint positions) of the source interaction. The third row visualizes the top-5 HOI edge attention weights (inter-subgraphs) aggregated from multiple interaction types, with lines connecting human joints to object part bounding box centers. The transition from Frame #0 to Frame $\#T$ demonstrates how the topological focus dynamically shifts as the sitting motion progresses.

2) STRIG Former Attention Visualization:: As illustrated in Fig. 4, we visualize the attention weights extracted by the STRIG Former. The top two rows display the top-3 nodes with the highest attention weights within the object subgraph and human subgraph of the source interaction's STRIG, corresponding to the red and blue nodes, respectively. The red nodes denote the center positions of the object part bounding boxes, while the blue nodes indicate the specific human joint positions. During the sitting motion from left (Frame #0) to right (Frame $\#T$), the local attention within the object subgraph consistently focuses on the features of the footrest and seat. Additionally, the attention weight naturally shifts from the left armrest to the right armrest. For the human subgraph, the weights transition logically from the weight-bearing left ankle, knee, and hand at Frame #0 to the pelvis and both ankles at Frame $\#T$, reflecting the physical motion phase. Furthermore, the HOI edge attention (inter-subgraphs) is aggregated from the four interaction edge types ($\mathcal{E}_{dist}, \mathcal{E}_{rel}, \mathcal{E}_{touch}, \mathcal{E}_{tow}$). We visualize the top-5 edges with the highest attention weights in the third row, with each edge connecting human joints to the centers of the object part bounding boxes. At Frame #0, the left hand and knee exhibit stronger interactive attention toward the left armrest and seat, while both ankles primarily focus on interacting with the lower structure of the chair (i.e., the base and footrest). By Frame $\#T$, the left and right hips, alongside the left knee, shift their interactive focus to the seat, while both ankles maintain strong attention on the footrest. These visualizations prove that the STRIG Former effectively captures complex internal and interactive topologies.

3) Other HOI Interactions With Multi-View Visualizations:: While our previous results primarily focused on chair-like sitting and seat-affording interactions, we further demonstrate the robustness of our method across a broader range of interactions. As illustrated in Fig. 5, we provide comprehensive multi-view visualizations (Left, Top, Right, and Back) of these transferred interactions. These multi-view results confirm that the generated motions consistently maintain strict motion similarity with the source sequences. This validates our method's capacity to effectively capture the holistic 3D topology, thereby ensuring physical plausibility from any observation view.

C. Comparisons

1) Comparison Methods:: We perform comparisons with both conventional and SOTA methods, including the surface-based method Tink [2], the spatial-correspondence-based method NDF [1], and the hybrid method SSCF [3]. SSCF operates on target objects with randomly assigned rotations, while Tink and NDF ignore the rotation of the target objects. To ensure a fair comparison, all methods are provided with the same target point cloud inputs that are properly aligned with the ground-truth orientations.

2) Protocol:: We randomly select 20 interactions from the CHAIRS dataset as source interactions, and 200 chairs from ShapeNet are used as target objects. The target objects are sampled from the test set, which has not been exposed to the network during training.

3) Evaluation Metrics:: For the HOI motion transfer task, we focus on the consistency of the motion sequence before and after transfer, the quality of human-object interactions, and the temporal coherence of the interaction. Inspired by SSCF, we compute the Joint Metric (JM) to comprehensively assess the SMPLX joints SO(3) geodesic distance [61] and the HOI penetration depth, weighted by $\lambda = 3$ and $\mu = 1.5$:

$$\mathrm{JM} = \mathbb{E}\left[\exp\left(-\lambda\, \mathbb{E}_j\left[d_{\mathrm{SO}(3)}(\mathbf{R}_j^{\mathtt{src}}, \mathbf{R}_j^{\mathtt{tgt}})\right] - \mu\, \mathcal{L}_{\mathrm{pen}}\right)\right] \quad (13)$$

To evaluate interaction consistency, we define the valid interactive region $\mathcal{V}_{int}^{\mathcal{O}}$ on the target object and the corresponding human region $\mathcal{V}_{int}^{\mathcal{H}}$ based purely on geometric normal compatibility. If the angle between a human vertex normal and a target object vertex normal is obtuse (indicating facing surfaces), these vertices are included in the interactive regions.

The Intersection over Union (IoU) computes the contact between the actual contact region $\mathcal{V}_{\mathrm{act}}$ and the interactive region in target objects as (14), and the Temporal Average Distance (TAD) measures the average distance from $\mathcal{V}_{\mathrm{int}}$ to the closest point on the human mesh $\mathcal{M}^{\mathcal{H}}$ as (15). A higher IoU and a smaller TAD indicate a tighter human-object interaction.

$$\mathrm{IoU} = \frac{\left|\mathcal{V}_{\mathrm{act}}^{\mathcal{O}} \cap \mathcal{V}_{\mathrm{int}}^{\mathcal{O}}\right|}{\left|\mathcal{V}_{\mathrm{act}}^{\mathcal{O}} \cup \mathcal{V}_{\mathrm{int}}^{\mathcal{O}}\right|} \quad (14)$$

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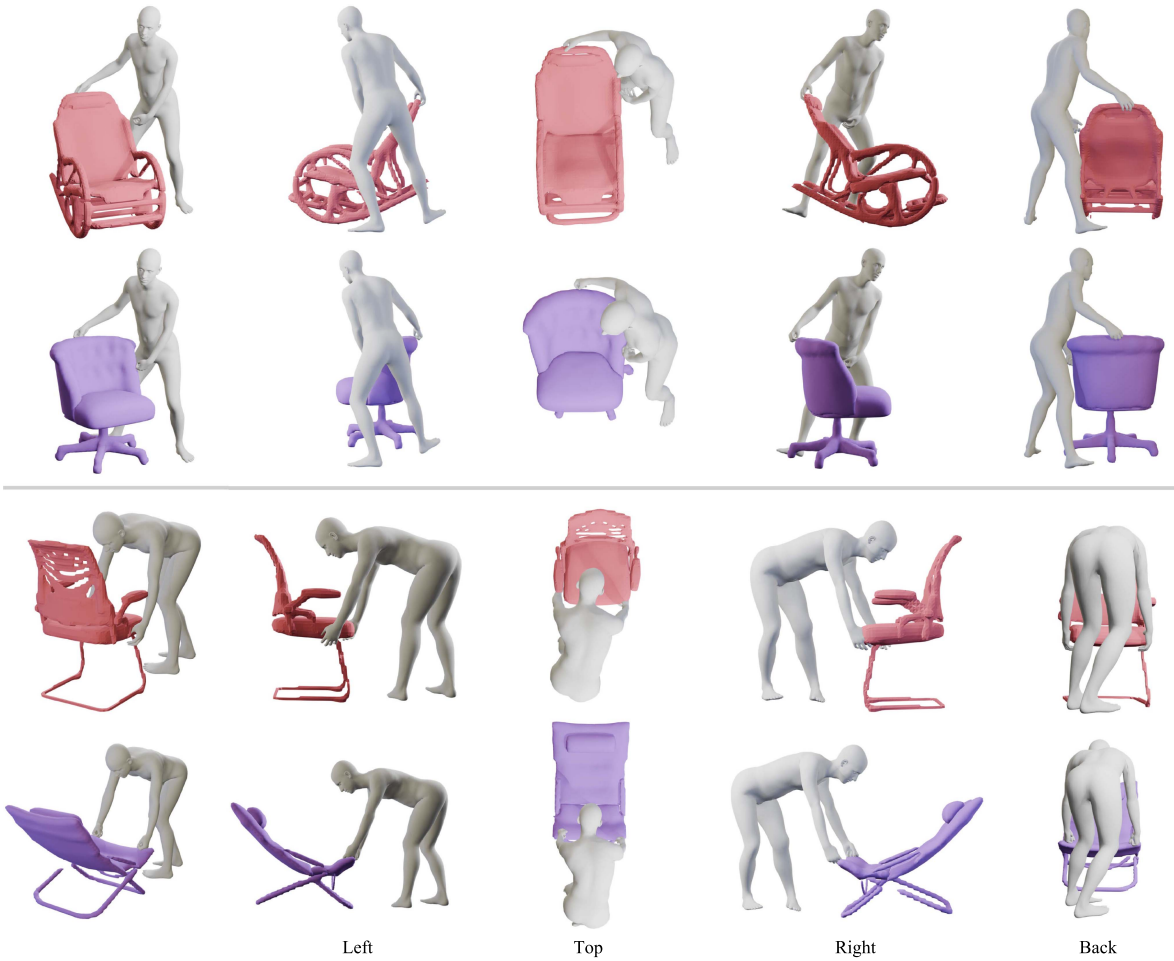


Fig. 5. Other HOI interactions with multi-view visualizations. We visualize two interaction transfer cases across diverse target geometries from multiple views. The multi-view visualizations demonstrate precise motion transfer and the absence of interpenetration, indicating improved 3D spatial consistency and physical plausibility of our method.

$$\mathrm{TAD} = \min_{t\in\{1,\ldots,T\}} \frac{1}{|\mathcal{V}^t_{\mathrm{int}}|} \sum_{\mathbf{v}\in\mathcal{V}^t_{\mathrm{int}}} \mathrm{dist}\big(\mathbf{v},\, \mathcal{M}^{\mathcal{H},t}\big) \tag{15}$$

We also use Contact Quality (CQ) to represent the precision of the interaction as (16).

$$\mathrm{CQ} = \frac{|\mathcal{V}^{\mathcal{O}}_{\mathrm{act}} \cap \mathcal{V}^{\mathcal{O}}_{\text{no-pen}}|}{|\mathcal{V}^{\mathcal{O}}_{\mathrm{act}}| + \varepsilon} \times \left(1 - \frac{|\mathcal{V}^{\mathcal{H}}_{\mathrm{act}} \setminus \mathcal{V}^{\mathcal{H}}_{\mathrm{int}}|}{|\mathcal{V}^{\mathcal{H}}_{\mathrm{act}}| + \varepsilon}\right) \tag{16}$$

Additionally, we use Contact Temporal Consistency (CTC) to measure the stability and coherence of the contact region throughout the HOI sequence as (17).

$$\mathrm{CTC} = \frac{1}{2(T-1)} \sum_{t=1}^{T-1} \sum_{\mathcal{X}\in\{\mathcal{H},\mathcal{O}\}} \left(1 - \frac{|\mathcal{V}^{\mathcal{X},(t)}_{\mathrm{act}} \setminus \mathcal{V}^{\mathcal{X},(t+1)}_{\mathrm{act}}|}{|\mathcal{V}^{\mathcal{X},(t)}_{\mathrm{act}}| + \varepsilon}\right) \tag{17}$$

For the human-chair interaction, we set a threshold of 4 cm. We also calculate the average transfer time in seconds for each method over a 10-frame HOI sequence.

4) Static Transfer Comparison:: Fig. 6 presents qualitative comparisons between our method and prior approaches on static HOI motion transfer. As shown in Fig. 6, each row illustrates the transfer of a source interaction to two different target objects. In the first two rows, where the subject sits on a chair and lightly rests a hand on the armrests while the target chair lacks them, our method better preserves the intent of the source interaction. The transferred poses remain consistent, with the hand naturally relaxing toward the seat, thereby ensuring physical plausibility and maintaining semantic fidelity. In the third and fourth rows, which involve more challenging sitting and leaning configurations, the spatial-correspondence-based method (NDF) fails to accurately align the spatial mapping between source and target surfaces, leading to misplaced or unstable contacts. The surface-based method (Tink) frequently exhibits penetration artifacts or erroneous contact regions due to limited geometric reasoning, while the hybrid method (SSCF) inherits the both drawbacks, resulting in degraded transfer quality. In contrast, our method preserves the relative spatial relationships from the source interaction, producing more coherent contact regions with significantly reduced penetration artifacts. For the final rows, our method generalizes more effectively to target objects with substantial geometric and topological variations, such as transferring chair-related motions to benches or ladders. Even under large geometric discrepancies, it generates transferred results that remain physically plausible and semantically consistent with the source interaction.

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Fig. 6. Qualitative comparison on static HOI motion transfer. Each row shows a source interaction (pink) transferred to target objects (purple).

5) Sequence Transfer Comparison:: Fig. 7 shows a qualitative comparison of HOI sequence transfer, where five uniformly sampled frames from each 10-frame source and transferred target sequence are instructed. Each row shows the temporal motion sequence from left to right. While NDF roughly transforms the motion to the target object, it often generates inconsistent motions, with the human partially floating above the surface and exhibiting noticeable temporal discontinuities. Tink and SSCF yield more plausible results, however, when transferring across a large topological change of the object (from an office chair to a long bench), they still generate unnatural poses, particularly in the pose of the legs. Besides, the foot trajectories in Tink are unstable and non-smooth. In contrast, our method better preserves the source interaction and achieves higher sequence stability and motion transfer quality, producing temporally smooth and physically more plausible HOI motion sequences. The last column of Tables I, II, III show the quantitative comparison between our approach and the SOTA method based on CTC

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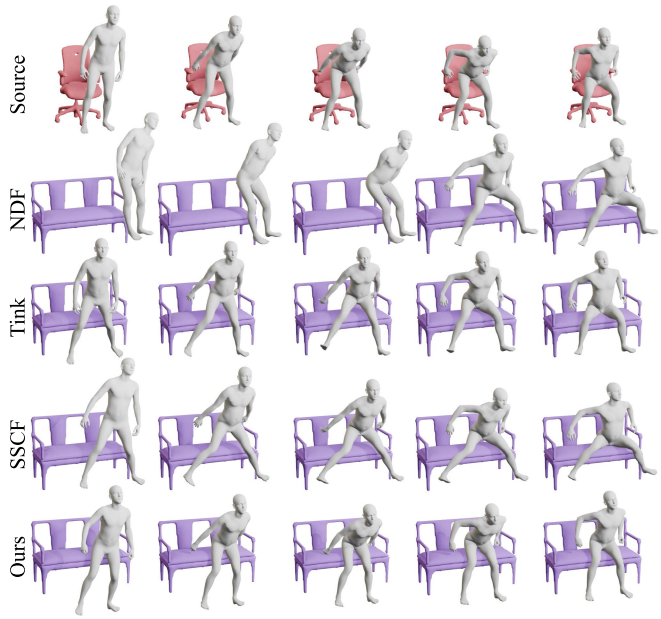


Fig. 7. Qualitative comparison on HOI motion sequence transfer. Each row shows a motion sequence over time (from left to right).

TABLE I
QUANTITATIVE QUALITY COMPARISON ON *SHAPENET*

Method	JM ↑	IoU ↑	TAD ↓	CQ ↑	CTC ↑
Motion-Copy	0.4267	0.2964	0.0757	0.0302	0.8279
NDF [1]	0.3845	0.1188	0.1960	0.0273	0.5021
Tink [2]	0.4029	0.0974	0.0742	0.1081	0.7869
SSCF [3]	0.4697	0.3023	**0.0397**	0.0826	0.7904
Ours	**0.5536**	**0.3206**	0.0425	**0.2540**	**0.8311**

TABLE II
QUANTITATIVE QUALITY COMPARISON ON *CHAIRS*

Method	JM ↑	IoU ↑	TAD ↓	CQ ↑	CTC ↑
Motion-Copy	0.6414	0.3129	0.0420	0.1498	0.8382
NDF [1]	0.2964	0.2774	0.1815	0.0829	0.6122
Tink [2]	0.3516	0.2588	0.0919	0.1558	0.7909
SSCF [3]	0.6178	0.3274	0.0412	0.1755	0.8353
Ours	**0.6787**	**0.3857**	**0.0353**	**0.2461**	**0.8561**

TABLE III
QUANTITATIVE QUALITY COMPARISON ON *PARK*

Method	JM ↑	IoU ↑	TAD ↓	CQ ↑	CTC ↑
Motion-Copy	0.3016	0.1291	1.3580	0.1070	0.7691
NDF [1]	0.2733	0.0479	2.4150	0.0554	0.5363
Tink [2]	0.4704	0.0847	1.4267	0.0780	0.6011
SSCF [3]	0.3142	0.0657	1.7736	0.0435	0.7211
Ours	**0.5385**	**0.2016**	**0.3484**	**0.2079**	**0.7863**

metric. Our method achieves the temporal consistency across all benchmarks, which also confirms that our method generates temporally smooth HOI motion sequences.

6) Motion-Copy Comparison:: To further quantify the necessity of interaction adaptation, we introduce a *Motion-Copy* baseline. This baseline directly applies the original source motion to the target object without any geometric adaptation. As shown in Table I, II, III, while *Motion-Copy* can achieve moderate metric scores on datasets with high geometric similarity between source and target objects (e.g., CHAIRS), it exhibits severe limitations in ensuring physical plausibility and precise contact. Specifically, our method demonstrates a substantial improvement in Contact Quality over the *Motion-Copy* baseline across all datasets. These results prove our method performances result from spatial-temporal adaptation, not inherent geometric similarity.

TABLE IV
QUANTITATIVE EFFICIENCY COMPARISON

Method	NDF [1]	Tink [2]	SSCF [3]	Ours
Time ↓	186.07s	287.25s	34.10s	**5.30s**

7) Quantitative Comparison:: Tables I, II, III, and IV present quantitative comparisons between our approach and the SOTA methods on the ShapeNet, CHAIRS, and PARK datasets. Our method achieves the highest JM and CTC across all benchmarks, which demonstrate the transferred motions better preserve both interaction integrity and temporal coherence. On the ShapeNet (Table I) and CHAIRS (Table II)—where the target object topologies are relatively similar—our approach produces tighter and more physically plausible interactions. In particular, the higher CQ highlights the improved precision and reliability of human–object contact. In the more challenging PARK dataset (Table III), complex object geometries introduce substantial variations in interaction topology, causing SSCF's pre-trained deformation field (RDIF) to struggle with establishing valid correspondences [3]. Despite these challenges, our method still yields substantial improvements across all metrics, underscoring its robustness and adaptability.

Furthermore, as shown in Table IV, our approach demonstrates remarkable computational efficiency, benefiting from processing motion sequences as holistic model inputs rather than performing frame-by-frame transfer. Overall, these results demonstrate that our method achieves the best balance among penetration control, contact precision, and temporal stability.

D. Ablations

We conduct ablation studies to evaluate the effectiveness of the STRIG representation and condition strategy within the diffusion model on the CHAIRS dataset, as summarized in Table V(a). When the STRIG representation is removed (w/o STRIG), the model is conditioned only on the SMPLX motion sequence and the source object. This leads to a clear degradation in spatial realism and temporal coherence, indicating that the model struggles to maintain consistent human–object contact without the explicit spatial-temporal relational guidance provided by STRIG. Introducing STRIG embeddings via concatenation with the noised target motion sequence (embed cat) significantly improves performance across all metrics. These gains validate the importance of explicitly encoding interaction relations for accurate and stable HOI transfer. Our full model applies attention-based conditioning (atten), where STRIG serves as a control attention signal within the diffusion decoder. This formulation further enhances both spatial and temporal consistency, achieving the best overall balance among all metrics. These results demonstrate that STRIG is essential for modeling

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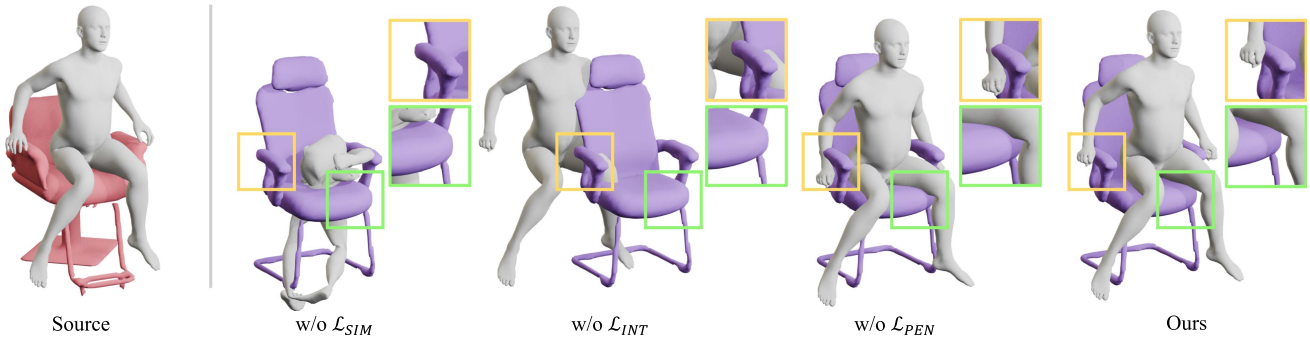


Fig. 8. Ablation studies of spatial relation optimization strategy. From left to right: source interaction, results without $\mathcal{L}_{SIM}$, $\mathcal{L}_{INT}$, $\mathcal{L}_{PEN}$, and our full model.

TABLE V
ABLATION STUDY ON DIFFERENT COMPONENTS OF OUR METHOD. WE EVALUATE (A) STRIG REPRESENTATION AND CONDITION STRATEGY, (B) STRIG FORMER ATTENTIONS, (C) SPATIAL RELATION OPTIMIZATION STRATEGY, AND (D) DIFFUSION STEPS.

	JM ↑	IoU ↑	TAD ↓	CQ ↑	CTC ↑
(a) STRIG Representation and Condition Strategy					
w/o STRIG	0.5741	0.3129	0.1060	0.1532	0.5692
embed cat	0.6261	0.3524	0.0420	0.2209	0.8155
Ours (atten)	**0.6787**	**0.3857**	**0.0353**	**0.2461**	**0.8561**
(b) STRIG Former Attentions					
w/o LA	0.6286	0.3718	0.0435	0.2207	0.8286
w/o IA	0.6497	0.3503	0.0440	0.2214	0.8175
w/o GA	0.6456	0.3702	0.0410	0.2385	0.8513
w/ IA (only)	0.4636	0.2990	0.0582	0.1566	0.6212
w/ GA (only)	0.4610	0.2103	0.0622	0.1461	0.6152
Ours	**0.6787**	**0.3857**	**0.0353**	**0.2461**	**0.8561**
(c) Spatial Relation Optimization Strategy					
w/o $\mathcal{L}_{\text{SIM}}$	0.0009	0.0930	0.0616	0.0918	0.6729
w/o $\mathcal{L}_{\text{INT}}$	0.6465	0.1428	0.1045	0.2180	0.7392
w/o $\mathcal{L}_{\text{PEN}}$	0.6566	0.3643	0.0359	0.2305	**0.8684**
Ours	**0.6787**	**0.3857**	**0.0353**	**0.2461**	0.8561
(d) Diffusion Steps					
$Step = 1$	0.5384	0.2892	0.0714	0.2140	0.7604
$Step = 2$	0.6408	0.3438	0.0431	0.2329	0.8381
$Step = 5$	0.6626	0.3714	0.0426	0.2396	0.8550
$Step = 20$	0.6227	0.3828	0.0381	0.2459	**0.8651**
$Step = 10$ (Ours)	**0.6787**	**0.3857**	**0.0353**	**0.2461**	0.8561

the spatial-temporal relational constraints of the source interaction, and that attention-based conditioning provides a more effective means of leveraging this representation, yielding fewer penetration artifacts, and stronger temporal coherence.

We then evaluate the individual contributions of the attention mechanisms (Local Attention LA, Interactive Attention IA and Global Aggregation GA) within the STRIG Former, detailed in Table V(b). Removing Local Attention (w/o LA) significantly degrades the JM, demonstrating that capturing relative local spatial features within the human and object subgraphs is critical for maintaining overall pose accuracy. Excluding Interactive Attention (w/o IA) results in the most severe drops in IoU and CTC among the attention ablations. This confirms that IA is essential for explicitly modeling the cross-domain spatial-temporal interaction edges ($\mathcal{E}_{\mathcal{HO}}$); without it, the model struggles to generate precise and temporally stable human-object contacts. Finally, omitting Global Aggregation (w/o GA) reduces performance across the board, underscoring its necessity in holistically integrating heterogeneous graph features into a unified latent representation. These prove that these attention mechanisms are highly complementary and crucial for extracting a robust spatial-temporal relational representation. It is worth noting that removing IA yields worse IoU, TAD, CQ, and CTC than removing GA, where the structural context of $\mathcal{G}_{\mathcal{O}}$ is fused. This suggests that the attributed HOI edges $\mathcal{E}_{\mathcal{HO}}$ provide more effective cues for human-object adaptation. Although w/o IA obtains a slightly higher JM than w/o GA, this indicates that removing $\mathcal{E}_{\mathcal{HO}}$ may preserve pose similarity, but at the cost of weaker object adaptation and degraded contact quality.

We further evaluate the contribution of each component in the proposed spatial relation optimization, as summarized in Table V(c) and Fig. 8. Removing the pose similarity term $\mathcal{L}_{SIM}$ leads to non-convergent motion generation, underscoring its necessity for maintaining coherent body dynamics. Excluding the interaction loss $\mathcal{L}_{INT}$ produces unstable grounding and inaccurate contacts, highlighting its role in preserving spatial consistency. Omitting the penetration term $\mathcal{L}_{PEN}$ yields plausible but interpenetrated results, showing its importance for fine-grained clearance. The reason why the CTC score increases without $\mathcal{L}_{PEN}$ is that the CTC only measures the temporal consistency of contact and ignores physical realism. When part of the target object lies along the trajectory of the original motion sequence, the model must generate a new motion trajectory to avoid the obstacle, which inevitably reduces the temporal consistency. The full optimization achieves the most stable and physically consistent outcomes, demonstrating that these complementary objectives jointly ensure realistic, contact-aware, and temporally coherent HOI motion transfer.

Finally, we investigate the impact of the diffusion steps on motion transfer quality, as summarized in Table V(d). Setting the process to a single step ($Step = 1$) causes a severe degradation across all metrics. In this configuration, the model bypasses the iterative denoising process and attempts to output the transferred sequence in a single forward pass. This demonstrates that one step is insufficient to resolve the complex spatial-temporal mapping between the source interaction and the novel target object. As the steps increase from 1 to 10, both physical realism and contact precision consistently improve. While increasing the steps to 20 improves temporal stability and coherence (CTC) by generating motions closer to the source trajectory, it simultaneously worsens interpenetration (like JM) on target objects exhibiting large topological differences. Thus, we determine $Step = 10$ to be the optimal configuration.

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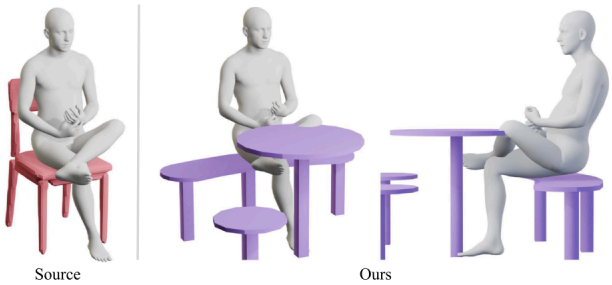


Fig. 9. Failure case of our method. When an obstacle, such as a table, is located in front of the target object, the transferred motion fails to perform collision-aware avoidance.

V. Conclusion

In this paper, we have presented a spatial–temporal relation guided motion transfer framework that leverages the proposed STRIG representation and diffusion-based optimization to achieve spatially and temporally consistent and physically plausible HOI motion generation. Extensive experiments demonstrate that our method significantly outperforms existing approaches in motion transfer quality, performance, and sequence stability across diverse object geometries and interaction topologies.

Limitations and Future Work: First, its reliance on the macroscopic SMPLX pose space restricts fine-grained inter-joint reasoning and lacks explicit optimization for complex hand-joint articulations. Second, although our spatial relation optimization effectively mitigates local collisions via penetration penalties $\mathcal{L}_{PEN}$, it lacks explicit, long-horizon path planning capabilities. Consequently, as shown in Fig. 9, the model may struggle to generate obstacle-aware transferred motions when large external obstacles, such as a table, are placed in front of the target chair, resulting in severe interpenetration between the human body and the obstacle.

Furthermore, the current framework cannot accommodate articulated or dynamic objects. Future work will address these limitations by integrating global obstacle-aware path planning, fine-grained inter-joint and hand-joint relational reasoning, and object-specific hand-object contact modeling to better support interactions that require precise finger articulation and grasp adaptation. In addition, we will incorporate deeper semantic and functional modeling to enhance cross-category generalization and robustness, particularly for sophisticated human-object interactions involving articulated and dynamic objects. Finally, our method mainly focus on chair-like and support-affording interactions. We will explore a broader range of interaction categories with substantially different affordances.

References

[1] A. Simeonov et al., "Neural descriptor fields: Se (3)-equivariant object representations for manipulation," in *Proc. Int. Conf. Robot. Automat.*, 2022, pp. 6394–6400.
[2] L. Yang et al., "OakInk: A large-scale knowledge repository for understanding hand-object interaction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 20953–20962.
[3] Z. Huang, H. Xu, H. Huang, C. Ma, H. Huang, and R. Hu, "Spatial and surface correspondence field for interaction transfer," *ACM Trans. Graph.*, vol. 43, no. 4, pp. 1–12, 2024.
[4] K. Sui, A. Ghosh, I. Hwang, B. Zhou, J. Wang, and C. Guo, "A survey on human interaction motion generation," *Int. J. Comput. Vis.*, vol. 134, no. 3, pp. 1–13, 2026.
[5] W. Zhu et al., "Human motion generation: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 4, pp. 2430–2449, Apr. 2024.
[6] F.-A. Croitoru, V. Hondru, R. T. Ionescu, and M. Shah, "Diffusion models in vision: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 9, pp. 10850–10869, Sep. 2023.
[7] K. Aberman, P. Li, D. Lischinski, O. Sorkine-Hornung, D. Cohen-Or, and B. Chen, "Skeleton-aware networks for deep motion retargeting," *ACM Trans. Graph.*, vol. 39, no. 4, pp. 62–1, 2020.
[8] Z. Ye, J.-W. Liu, J. Jia, S. Sun, and M. Z. Shou, "Skinned motion retargeting with dense geometric interaction perception," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 125907–125934.
[9] J. Zhang et al., "Skinned motion retargeting with residual perception of motion semantics & geometry," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 13864–13872.
[10] J. Zhang, Z. Tu, J. Weng, J. Yuan, and B. Du, "A modular neural motion retargeting system decoupling skeleton and shape perception," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 10, pp. 6889–6904, Oct. 2024.
[11] H. Zhang et al., "Semantics-aware motion retargeting with vision-language models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2024, pp. 2155–2164.
[12] J. Kim, Y. Seol, and T. Kwon, "Interactive multi-character motion retargeting," *Comput. Animation Virtual Worlds*, vol. 32, no. 3-4, 2021, Art. no. e2015.
[13] Y. Zhang, D. Gopinath, Y. Ye, J. Hodgins, G. Turk, and J. Won, "Simulation and retargeting of complex multi-character interactions," in *Proc. ACM SIGGRAPH Conf. Proc.*, 2023, pp. 1–11.
[14] Y. Liu, C. Zhang, R. Xing, B. Tang, B. Yang, and L. Yi, "Core4D: A 4D human-object-human interaction dataset for collaborative object rearrangement," in *Proc. Comput. Vis. Pattern Recognit. Conf.*, 2025, pp. 1769–1782.
[15] D. Rodriguez and S. Behnke, "Transferring category-based functional grasping skills by latent space non-rigid registration," *IEEE Robot. Autom. Lett.*, vol. 3, no. 3, pp. 2662–2669, Jul. 2018.
[16] R. Wu, T. Zhu, W. Peng, J. Hang, and Y. Sun, "Functional grasp transfer across a category of objects from only one labeled instance," *IEEE Robot. Autom. Lett.*, vol. 8, no. 5, pp. 2748–2755, May 2023.
[17] Y. Kim, H. Park, S. Bang, and S.-H. Lee, "Retargeting human-object interaction to virtual avatars," *IEEE Trans. Vis. Comput. Graphics*, vol. 22, no. 11, pp. 2405–2412, Nov. 2016.
[18] S. Choi, S. Hong, K. Cho, C. Kim, and J. Noh, "Online avatar motion adaptation to morphologically-similar spaces," *Comput. Graph. Forum*, vol. 42, pp. 13–24, 2023.
[19] D. H. Le et al., "One diffusion to generate them all," in *Proc. Comput. Vis. Pattern Recognit. Conf.*, 2025, pp. 2671–2682.
[20] J. Schusterbauer, M. Gui, F. Fundel, and B. Ommer, "Diff2Flow: Training flow matching models via diffusion model alignment," in *Proc. Comput. Vis. Pattern Recognit. Conf.*, 2025, pp. 28347–28357.
[21] K. Sun, Z. Chen, X. Lin, X. Sun, H. Liu, and R. Ji, "Conditional diffusion models for camouflaged and salient object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 4, pp. 2833–2848, Apr. 2025.
[22] Y. Li, Z. Liu, B. Benes, X. Zhang, and J. Guo, "SVDTree: Semantic voxel diffusion for single image tree reconstruction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 4692–4702.
[23] H. Pi et al., "Motion-2-to-3: Leveraging 2D motion data to boost 3D motion generation," 2024, *arXiv:2412.13111*.
[24] J. Tang, Y. Nie, L. Markhasin, A. Dai, J. Thies, and M. Nießner, "DiffuScene: Denoising diffusion models for generative indoor scene synthesis," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 20507–20518.
[25] M. Xu et al., "Energy-based diffusion language models for text generation," in *Proc. Int. Conf. Learn. Representations*, 2025, pp. 33769–33789.
[26] B. Xia et al., "DiffI2I: Efficient diffusion model for image-to-image translation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 3, pp. 1578–1593, Mar. 2025.
[27] Y. Song and S. Ermon, "Improved techniques for training score-based generative models," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 12438–12448.
[28] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 6840–6851.
[29] M. Zhang et al., "MotionDiffuse: Text-driven human motion generation with diffusion model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 6, pp. 4115–4128, Jun. 2024.

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[30] G. Tevet, S. Raab, B. Gordon, Y. Shafir, D. Cohen-or, and A. H. Bermano, "Human motion diffusion model," in *Proc. 11th Int. Conf. Learn. Representations*, 2023. [Online]. Available: https://openreview.net/forum?id=SJ1kSyO2jwu
[31] J. Kim, J. Kim, and S. Choi, "FLAME: Free-form language-based motion synthesis & editing," in *Proc. AAAI Conf. Artif. Intell.*, 2023, pp. 8255–8263.
[32] S. K. Dwivedi et al., "InteractVLM: 3D interaction reasoning from 2D foundational models," in *Proc. Comput. Vis. Pattern Recognit. Conf.*, 2025, pp. 22605–22615.
[33] X. Xie, B. L. Bhatnagar, J. E. Lenssen, and G. Pons-Moll, "Template free reconstruction of human-object interaction with procedural interaction generation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 10003–10015.
[34] G. Moon, H. Choi, and K. M. Lee, "Accurate 3D hand pose estimation for whole-body 3D human mesh estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 2308–2317.
[35] J. Zhang, Y. Chen, Z. Wang, J. Yang, Y. Wang, and S. Huang, "InteractAnything: Zero-shot human object interaction synthesis via LLM feedback and object affordance parsing," in *Proc. Comput. Vis. Pattern Recognit. Conf.*, 2025, pp. 7015–7025.
[36] J. Kim, J. Kim, J. Na, and H. Joo, "ParaHome: Parameterizing everyday home activities towards 3D generative modeling of human-object interactions," in *Proc. Comput. Vis. Pattern Recognit. Conf.*, 2025, pp. 1816–1828.
[37] J. Li, J. Wu, and C. K. Liu, "Object motion guided human motion synthesis," *ACM Trans. Graph.*, vol. 42, no. 6, pp. 1–11, 2023.
[38] J. Li et al., "GENMO: A generalist model for human motion," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, pp. 11766–11776, 2025.
[39] Y. Zhang et al., "A plug-and-play physical motion restoration approach for in-the-wild high-difficulty motions," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, pp. 13281–13292, 2025.
[40] Z. Shen et al., "World-grounded human motion recovery via gravity-view coordinates," in *Proc. SIGGRAPH Asia Conf. Papers*, 2024, pp. 1–11.
[41] Y. Zhang et al., "DiffGrasp: Whole-body grasping synthesis guided by object motion using a diffusion model," in *Proc. AAAI Conf. Artif. Intell.*, 2025, pp. 10320–10328.
[42] N. Jiang et al., "Scaling up dynamic human-scene interaction modeling," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 1737–1747.
[43] M. Tanaka and K. Fujiwara, "Role-aware interaction generation from textual description," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 15999–16009.
[44] S. Xu, Z. Li, Y.-X. Wang, and L.-Y. Gui, "InterDiff: Generating 3D human-object interactions with physics-informed diffusion," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 14928–14940.
[45] G. Pavlakos et al., "Expressive body capture: 3D hands, face, and body from a single image," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 10975–10985.
[46] Y. Zhou, C. Barnes, J. Lu, J. Yang, and H. Li, "On the continuity of rotation representations in neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 5745–5753.
[47] C. R. Qi, L. Yi, H. Su, and L. J. Guibas, "PointNet : Deep hierarchical feature learning on point sets in a metric space," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 5105–5114.
[48] N. Jiang et al., "Full-body articulated human-object interaction," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 9365–9376.
[49] L. Xue et al., "ULIP: Learning a unified representation of language, images, and point clouds for 3D understanding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 1179–1189.
[50] G. Tevet et al., "CLoSD: Closing the loop between simulation and diffusion for multi-task character control," in *Proc. Int. Conf. Learn. Representations*, 2025, pp. 46506–46520.
[51] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 6000–6010.
[52] R. Ron, G. Tevet, H. Sawdayee, and A. H. Bermano, "HOIDiNi: Human-object interaction through diffusion noise optimization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2026, pp. 5739–5749.
[53] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Lio, and Y. Bengio, "Graph attention networks," 2017, *arXiv:1710.10903*.
[54] Y. Shi, Z. Huang, S. Feng, H. Zhong, W. Wang, and Y. Sun, "Masked label prediction: Unified message passing model for semi-supervised classification," in *Proc. 13th Int. Joint Conf. Artif. Intell.*, 2021, pp. 1548–1554.
[55] W. Hamilton, Z. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 1025–1035.
[56] K. Xu, C. Li, Y. Tian, T. Sonobe, K.-I. Kawarabayashi, and S. Jegelka, "Representation learning on graphs with jumping knowledge networks," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 5453–5462.
[57] A. X. Chang et al., "ShapeNet: An information-rich 3D model repository," 2015, *arXiv:1512.03012*.
[58] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," 2020, *arXiv:2010.02502*.
[59] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," 2017, *arXiv:1711.05101*.
[60] D. Morales-Brotons, T. Vogels, and H. Hendrikx, "Exponential moving average of weights in deep learning: Dynamics and benefits," 2024, *arXiv:2411.18704*.
[61] V. Surazhsky, T. Surazhsky, D. Kirsanov, S. J. Gortler, and H. Hoppe, "Fast exact and approximate geodesics on meshes," *ACM Trans. Graph.*, vol. 24, no. 3, pp. 553–560, 2005.



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